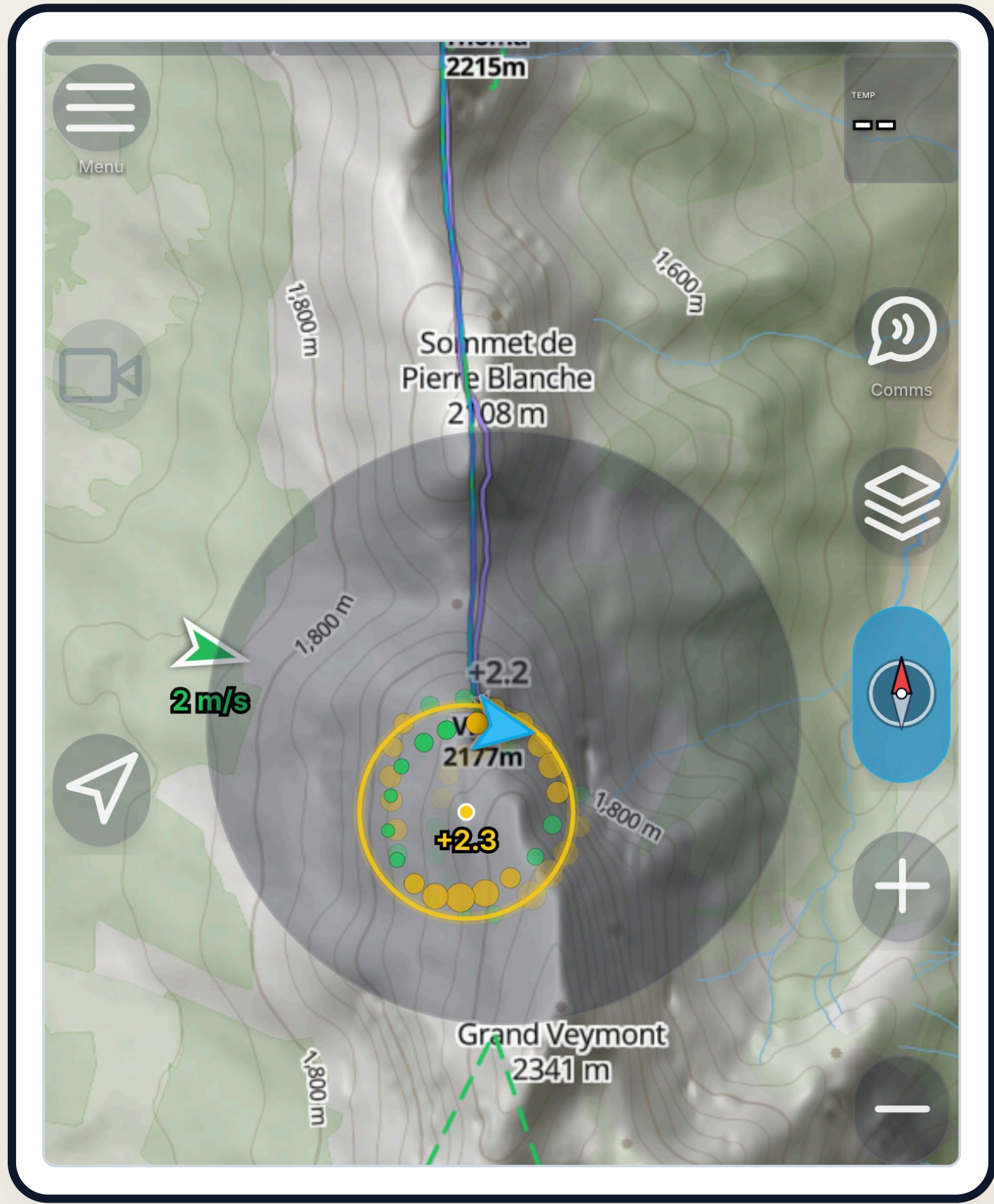


Algorithms of pgpilot

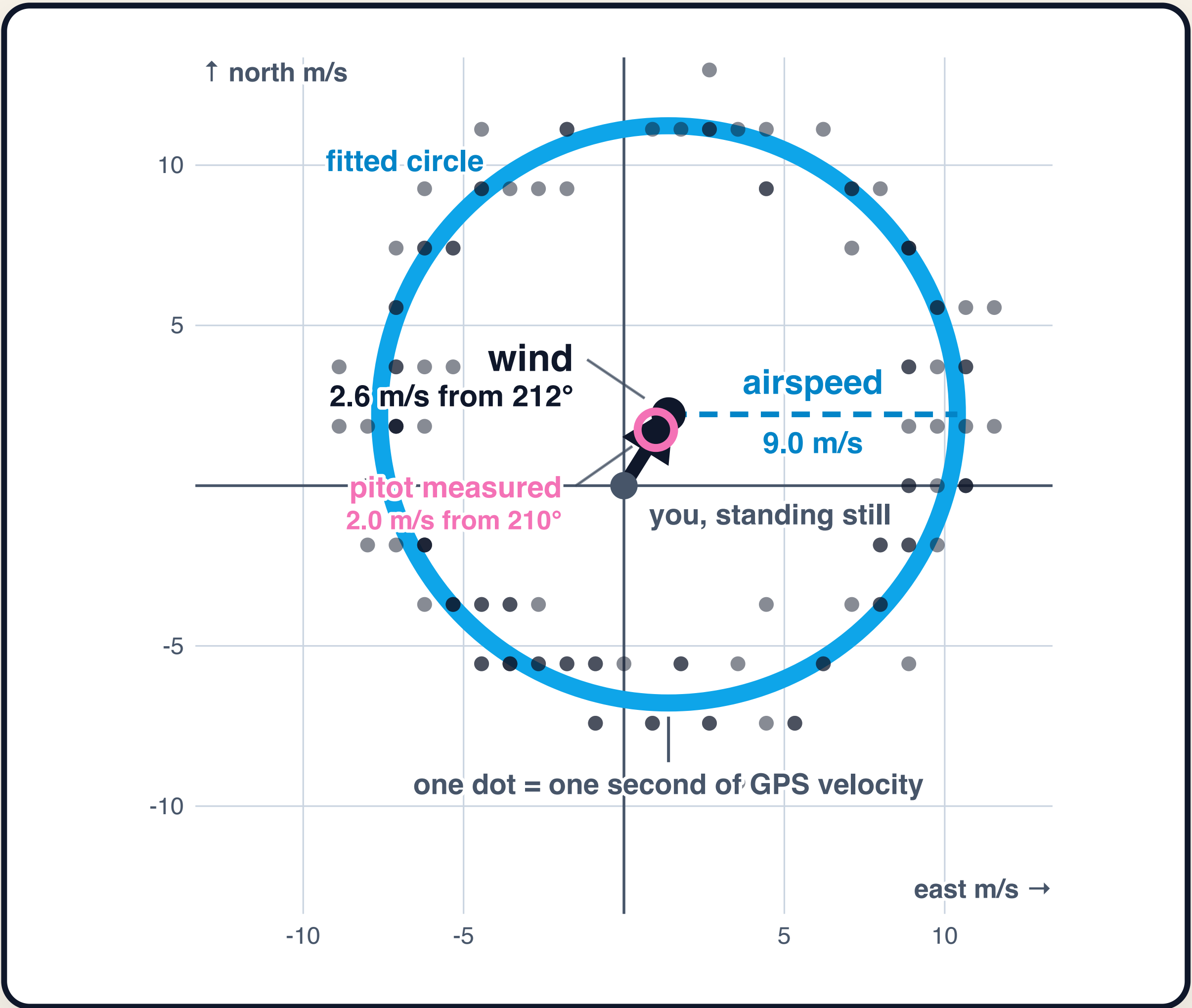
What runs on your phone, and what runs on the server.

01 Thermal centre



- Lift-weighted centre of the **last 2 minutes**
- **Drifted** with the wind
- Circling detected by a **4-state machine**

02 Wind estimator



- GPS velocities **fall on a circle**
- **Centre = wind, radius = airspeed**
- A **Kalman filter** fuses every circle, withdraws when unsure

613 labelled thermals · 89 h · blank time 15.4 to 6.6 s/h

vs pitot wind · circling 6 to 16°, speed ratio 1.0 · glide 21 to 34°

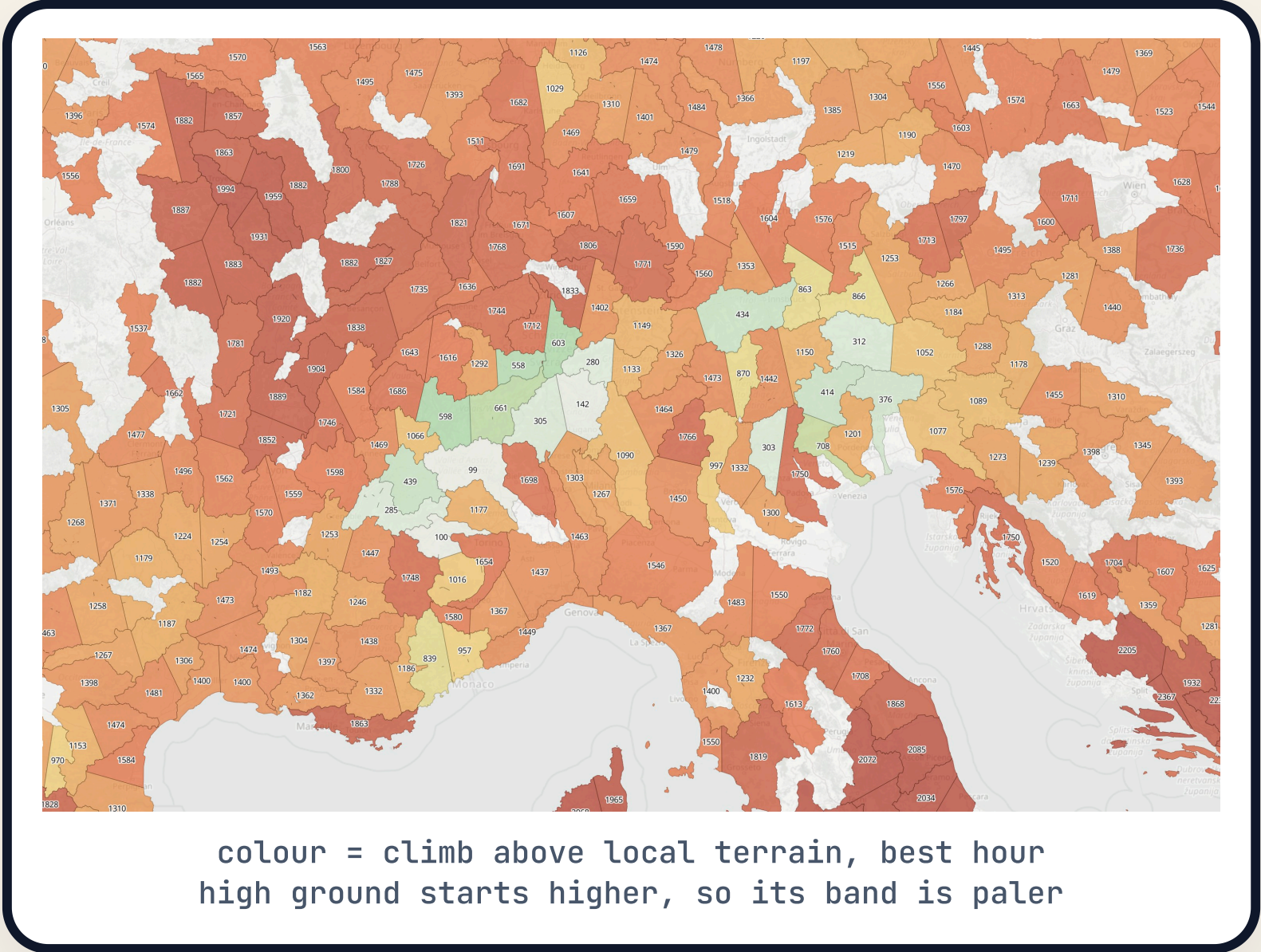
03 Point forecast



- **MEPS 2.5 km** in the Nordics, ICON-EU elsewhere, native levels
- A sounding per takeoff: **thermal top, cloud base, climb, wind by height**
- **Potential flight distance** per site per day

every flight keeps **the forecast it was flown on** · Hvittingfoss 13 Sept, 1 h 31 soaring, ICON-EU 05:00 run

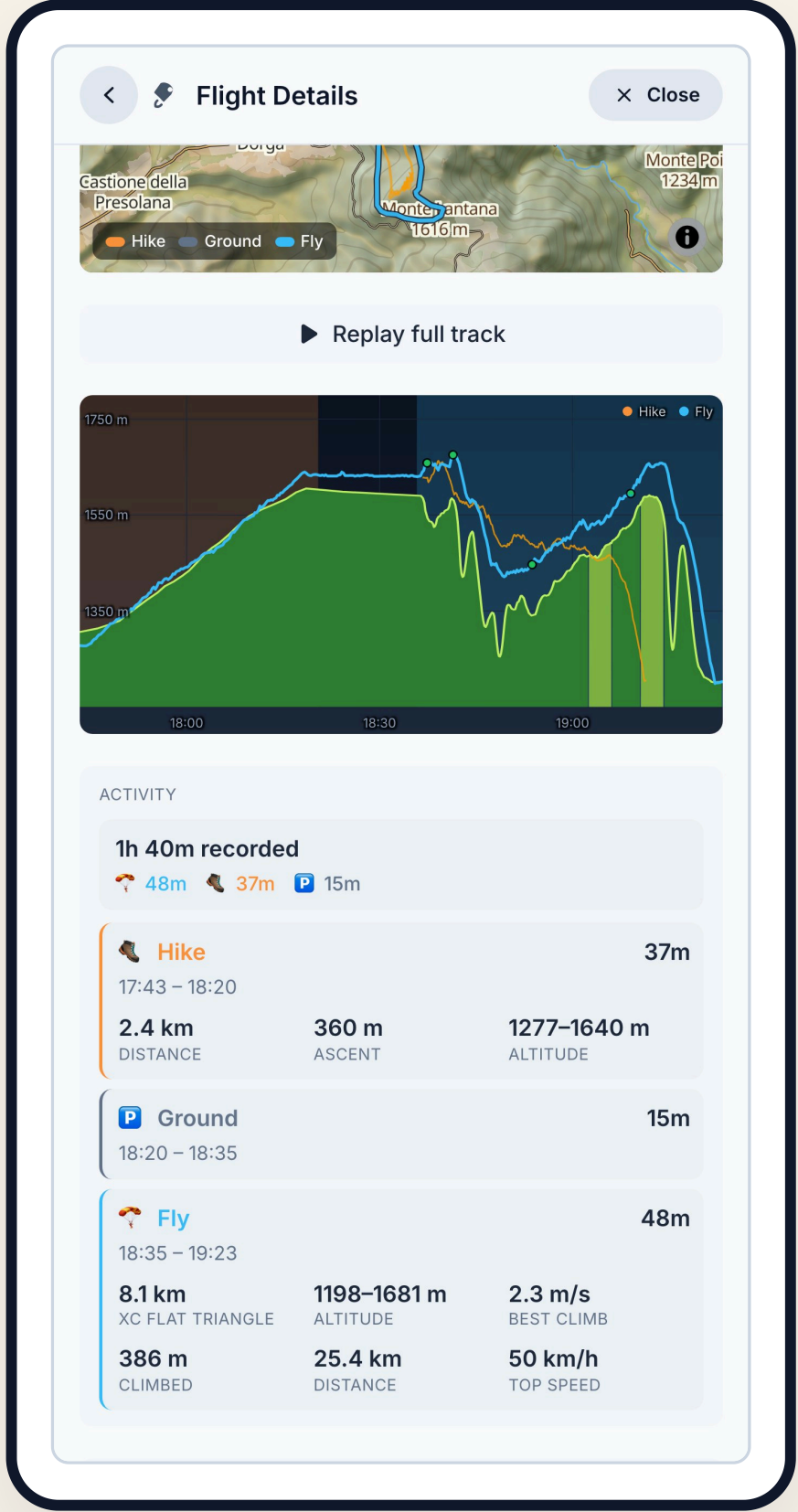
04 Regional forecast RESEARCH PROTOTYPE



- One forecast **per flying region**, not per grid point
- **1226 regions** from watersheds and **894 280 flights**
- Models: **ICON-EU, GFS, met.no**

ran Europe-wide once · matches mixed-layer theory within 4% · not yet validated

05 Activity segmentation



- One recording, several activities: **hike, fly, gondola, drive**
- **Viterbi decode** over speed, climb, height above ground, roads
- A **random forest** learned the costs from labelled tracks

194 hand-labelled recordings · 68 to 84% correct · zero missed flights

Measured against

9

flights with pitot wind

613

hand-labelled thermals

194

hand-labelled recordings

894 280

flights behind the regions

- **Next:** stations and pitot varios as truth · airspeed learned per wing · a thermal map from every synced flight

Scan me!



pgpilot.app
Sync your flights. Every flight sharpens the next estimate.